

API Testing Report — Nexora Workflows (Dev Environment)

Generated: 2026-06-07
API Base URL: <https://nexora-dev-api.vlinkhub.com>
Environment: Test (nexora-test DB — PostgreSQL at 157.230.44.126:5432)
Server: [Nexora.Web](#) (.NET 8.0) — SSO: <https://test-userservice.vlinkgroup.net>

Test Accounts Created

Email	Password	ProfileType	Role
merchant@nexora-dev.vlinkhub.com	Test@123456	Merchant	Business Owner
staff1@nexora-dev.vlinkhub.com	Test@123456	User	Staff (Anna Smith) — Active
staff2@nexora-dev.vlinkhub.com	Test@123456	User	Staff (Mike Johnson) — Active

Note: Admin accounts require SSO realm role `admin-portal` in Keycloak — not testable via API.

Summary

Flow	Steps	Status	Notes
Phase 0: Pre-flight	3	✔ Pass	API health 200, DB connected via psycopg2
Flow 1: Merchant Onboarding	10	✔ Pass	Business created, 3 touch points with QR codes
Flow 2: Staff via Invite (Anna)	11	✔ Pass	CashApp + Venmo configured
Flow 3: Staff via Invite (Mike)	6	✔ Pass	Zelle configured
Flow 4: Customer Tip Flow	10	✔ Pass	2 confirmed tips, 2 reviews (public + private), 1 skip
Flow 5: Merchant Dashboard	12	✔ Pass	Overview correct (totalAmount=15), reviews, tips, staff, touchpoints. 2 extra endpoints not implemented (404)
Flow 6: Public Endpoints	6	✔ Pass	Touch pages, banners, blog categories, staff tip history
Flow 7: Staff Management API	5	⚠ Partial	Toggle Active/Inactive ✔, Reorder ❌ 500 Error, Resend-invite ❌ 405
Flow 8: Security Tests	7	✔ Pass	All IDOR, payment handle exposure, token reuse, plan gate tests pass

Data IDs (Reference)

Resource	ID
Merchant UserProfile	fa9cd462-77d5-4023-84de-b4d1a3e40b0b
Staff1 (Anna) UserProfile	bcdcf39-0646-49ab-8037-56de2b53e445
Staff2 (Mike) UserProfile	6ebb45ae-44c5-4786-b2e1-c64f18459c1b

Resource	ID
Staff1 StaffProfile	bebbdfb9-43af-4c26-a056-70b0d35aeb91 (Code: N99MMUIQK)
Staff2 StaffProfile	b48d5643-4fde-4924-a17a-c2fa610b7c2e (Code: NYINFFK30)
Staff1 BusinessStaffLink	56f9cefe-5200-473e-9456-93ed293f6cfa
Staff2 BusinessStaffLink	ed492187-de43-47c4-910a-a2a4e8d6e80d
Business	e734f8db-5dad-4233-a7a6-039daccf92dd (Slug: nail-hub-dev)
Staff1 Invite ID	b28e5a3c-bff2-44c9-992e-59310e719867
Staff2 Invite ID	57f6b9dd-aaaa-452f-a624-e47dac5ebb78
TouchPoint: Table #1	0a7c35bd-5424-4eac-8e66-05c7c97e7982
TouchPoint: Front Desk	99b1295b-84d1-400e-9012-8cbdd4d35807
TouchPoint: Anna's Card	9eb64acc-3241-4a97-aaa7-49552ffaed85
Staff1 CashApp PaymentMethod	6686489a-dbef-4dcb-bfb6-b27b2a9c3006
Staff1 Venmo PaymentMethod	c02d8705-e4e5-46f0-b812-8e259ae74d4e
Staff2 Zelle PaymentMethod	d7e12e54-46e4-4d40-b4c5-4cbe0ee20255
Tip 1 (\$5 CashApp, Confirmed)	5ae07ccc-4e79-4e1b-890b-844856cbd597
Tip 2 (\$10 Venmo, Confirmed)	b5bd8888-0430-4706-8ddd-4af5ec5b2e84
Review 1 (5★, Public)	cdb6f10a-5cc3-422b-892e-2b034072dc01
Review 2 (2★, Private)	178db669-4d09-4591-a6fb-eb91c30a2786

Phase 0 — Pre-flight

Step 0.1 — Health Check

GET https://nexora-dev-api.vlinkhub.com/health

Response: HTTP 200

Step 0.2 — Swagger

GET https://nexora-dev-api.vlinkhub.com/swagger
GET https://nexora-dev-api.vlinkhub.com/swagger/index.html

Response: HTTP 404 — Swagger not accessible despite EnableSwagger: true in appsettings.Test.json. Likely disabled at reverse proxy/nginx level.

Step 0.3 — DB Connectivity

```
psycopg2.connect(host='157.230.44.126', port=5432, dbname='nexora-test',  
                  user='admin', password='...')
```

Response: Connected successfully.

Flow 1: Merchant Registration & Onboarding

Step 1.1 — Signup Merchant

POST <https://nexora-dev-api.vlinkhub.com/api/v1/authentication/signup>
Content-Type: application/json

```
{
  "email": "merchant@nexora-dev.vlinkhub.com",
  "password": "Test@123456",
  "confirmEmail": "merchant@nexora-dev.vlinkhub.com",
  "type": 1
}
```

Response: HTTP 201

```
{
  "isSuccess": true,
  "email": "merchant@nexora-dev.vlinkhub.com",
  "userId": "fa9cd462-77d5-4023-84de-b4d1a3e40b0b",
  "message": "Sign up new user successful",
  "errorCode": ""
}
```

Step 1.2 — Sign in Merchant

POST <https://nexora-dev-api.vlinkhub.com/api/v1/authentication/signin>
Content-Type: application/json

```
{"email": "merchant@nexora-dev.vlinkhub.com", "password": "Test@123456"}
```

Response: HTTP 200

```
{
  "accessToken": "<JWT_TOKEN>",
  "refreshToken": "<REFRESH_TOKEN>",
  "expiresIn": 604800
}
```

Step 1.3 — Get Current Profile

GET <https://nexora-dev-api.vlinkhub.com/api/v1/userprofile/me>
Authorization: Bearer <MERCHANT_TOKEN>

Response: HTTP 200

```
{
  "id": "fa9cd462-77d5-4023-84de-b4d1a3e40b0b",
  "email": "merchant@nexora-dev.vlinkhub.com",
  "userType": "Merchant",
  "status": "Active",
  "isEmailVerified": true,
  "country": "Vietnam",
  "referralCode": "752531CE"
}
```

Step 1.4 — Create Business

POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/business
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{
  "name": "Nail Hub Dev",
  "slug": "nail-hub-dev",
  "businessType": "Nail Salon",
  "description": "Dev environment nail salon for API testing",
  "address": "456 Test Ave",
  "city": "Los Angeles",
  "state": "CA",
  "phone": "3105559876",
  "website": "https://nailhubdev.example.com",
  "timeZone": "America/Los_Angeles"
}
```

Response: HTTP 201

```
{
  "businessId": "e734f8db-5dad-4233-a7a6-039daccf92dd",
  "slug": "nail-hub-dev"
}
```

Step 1.5 — Update Review Links

PUT https://nexora-dev-api.vlinkhub.com/api/v1/merchant/business/review-links
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{
  "googleReviewUrl": "https://g.co/nail-hub-dev",
  "yelpUrl": "https://yelp.com/biz/nail-hub-dev",
  "feedbackEmail": "merchant@nexora-dev.vlinkhub.com"
}
```

Response: HTTP 200 (*empty body*)

Step 1.6 — Invite Staff1 (Anna Smith)

POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/staff/invite
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{"invitedName": "Anna Smith", "invitedEmail": "staff1@nexora-dev.vlinkhub.com", "invitedPosition": "Nail Tech"}
```

Response: HTTP 201

```
{"inviteId": "b28e5a3c-bff2-44c9-992e-59310e719867"}
```

Token tür DB: -GQSTF2FAwME0rvinVZ1IftG5rL0_MKEtmGxuQT_FY80gwME1pfkM0hhE36aXtLv0I42VeYAmHT0SXJ-62D19g

Step 1.7 — Invite Staff2 (Mike Johnson)

POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/staff/invite
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{"invitedName": "Mike Johnson", "invitedEmail": "staff2@nexora-dev.vlinkhub.com", "invitedPosition": "Nail Artist"}
```

Response: HTTP 201

```
{"inviteId": "57f6b9dd-aca-452f-a624-e47dac5ebb78"}
```

Token tür DB: G9b-4S1zJWqvEVj145_UrCq-9-IcnfDh7IEbt_0y_N700RLqWg4ykj7UCkLu9Hk_a3vUnXLJa85Xff5X99gNQw

Step 1.8a — Create TouchPoint: Table #1

POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/touchpoints
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{"name": "Table #1", "type": 0}
```

Response: HTTP 201

```
{  
  "touchPointId": "0a7c35bd-5424-4eac-8e66-05c7c97e7982",  
  "qrImageUrl": "https://vlink-storages-test.sgp1.cdn.digitaloceanspaces.com/vlink-storages-test/Nexora/touchpoints/e734f8db-5dad-4233-  
}
```

Step 1.8b — Create TouchPoint: Front Desk

POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/touchpoints
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{"name": "Front Desk", "type": 1}
```

Response: HTTP 201

```
{  
  "touchPointId": "99b1295b-84d1-400e-9012-8cbbd4d35807",  
  "qrImageUrl": "https://vlink-storages-test.sgp1.cdn.digitaloceanspaces.com/vlink-storages-test/Nexora/touchpoints/e734f8db-5dad-4233-  
}
```

Step 1.8c — Create TouchPoint: Anna's Card (StaffCard) *(deferred after Flow 2 Step 2.4)* {#step-18c--create-touchpoint-annas-card-staffcard-deferred-after-flow-2-step-24 }

POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/touchpoints
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

```
{"name": "Anna's Card", "type": 3, "assignedStaffProfileId": "bebbdfb9-43af-4c26-a056-70b0d35aeb91"}
```

Response: HTTP 201

```
{
  "touchPointId": "9eb64acc-3241-4a97-aaa7-49552ffaed85",
  "qrImageUrl": "https://vlink-storages-test.sgp1.cdn.digitaloceanspaces.com/vlink-storages-test/Nexora/touchpoints/e734f8db-5dad-4233-
}
```

Note: Starter plan giới hạn 3 touch points.

Step 1.9 — Complete Onboarding

POST <https://nexora-dev-api.vlinkhub.com/api/v1/merchant/business/complete-onboarding>
Authorization: Bearer <MERCHANT_TOKEN>

Response: HTTP 200 — Business isPublic: true, onboardingStep: 5

Flow 2: Staff via Invite — Anna Smith (staff1@nexora-dev.vlinkhub.com)

Step 2.1 — Get Invite Info (Anonymous)

GET https://nexora-dev-api.vlinkhub.com/api/v1/staff/invite/-GQSTF2FAwmE0rvinVZ1IftG5rL0_MKEtmGxuQT_FY80gWME1pfkM0hhE36aXtLv0I42VeYAmH

Response: HTTP 200

```
{
  "invitedName": "Anna Smith",
  "invitedPosition": "Nail Tech",
  "businessName": "Nail Hub Dev"
}
```

Step 2.1b — Invalid/Expired Token Test

GET https://nexora-dev-api.vlinkhub.com/api/v1/staff/invite/INVALID_EXPIRED_TOKEN_12345

Response: HTTP 404

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.4",
  "title": "Not Found",
  "status": 404,
  "errorDetail": [{"message": "INVALID_EXPIRED_TOKEN_12345", "errorCode": "StaffInvite"}]
}
```

Note: Invalid/expired token returns 404 (plan expected 410 Gone). Behavior differs from US-13 spec.

Step 2.2 — Signup Staff1

POST <https://nexora-dev-api.vlinkhub.com/api/v1/authentication/signup>
Content-Type: application/json

```
{"email": "staff1@nexora-dev.vlinkhub.com", "password": "Test@123456", "confirmEmail": "staff1@nexora-dev.vlinkhub.com", "type": 0}
```

Response: HTTP 201

```
{"isSuccess": true, "userId": "bcdcf39-0646-49ab-8037-56de2b53e445"}
```

Step 2.3 — Sign in Staff1

POST https://nexora-dev-api.vlinkhub.com/api/v1/authentication/signin

Content-Type: application/json

```
{"email": "staff1@nexora-dev.vlinkhub.com", "password": "Test@123456"}
```

Response: HTTP 200 — JWT token returned

Step 2.4 — Accept Invite (Anonymous — no auth required)

POST https://nexora-dev-api.vlinkhub.com/api/v1/staff/invite/-GQSTF2FAwmE0rvinVZ1IftG5rl0_.../accept

Content-Type: application/json

```
{
  "displayName": "Anna Smith",
  "position": "Nail Tech",
  "bio": "Professional nail tech with 5 years experience",
  "photoUrl": "https://raw.githubusercontent.com/vlink-group/public-assets/refs/heads/main/the-nail-hub/DummyImage/Thumbnail/1.jpg"
}
```

Response: HTTP 204 (*StaffProfile + BusinessStaffLink created + 5 StaffPaymentMethod records*)

DB verification:

- StaffInvite.Status = Accepted 
- BusinessStaffLink.Status = Active 
- StaffProfile.Id = bebbdfb9-43af-4c26-a056-70b0d35aeb91 , StaffCode = N99MMUIQK

Step 2.5 — Update Staff Profile

PUT https://nexora-dev-api.vlinkhub.com/api/v1/staff/profile

Authorization: Bearer <STAFF1_TOKEN>

Response: HTTP 204

Step 2.6 — Get Payment Methods

GET https://nexora-dev-api.vlinkhub.com/api/v1/staff/payment-methods

Authorization: Bearer <STAFF1_TOKEN>

Response: HTTP 200 — 5 payment methods (Zelle, PayPal, Venmo, CashApp, AppleCash) pre-created with `isActive: false`

Step 2.7 — Configure CashApp (\$anna.smith)

PUT https://nexora-dev-api.vlinkhub.com/api/v1/staff/payment-methods/6686489a-dbef-4dcb-bfb6-b27b2a9c3006

Authorization: Bearer <STAFF1_TOKEN>

Content-Type: application/json

```
{"accountInfo": "$anna.smith", "imageUrl": null}
```

Response: HTTP 200 — `{"isActive": true, "isConfigured": true, "isConfigured": true}`

PATCH https://nexora-dev-api.vlinkhub.com/api/v1/staff/payment-methods/6686489a-dbef-4dcb-bfb6-b27b2a9c3006/toggle

Authorization: Bearer <STAFF1_TOKEN>

Response: HTTP 200 — `{"isActive": true}`

Step 2.8 — Configure Venmo (@anna.smith)

```
PUT .../payment-methods/c02d8705-e4e5-46f0-b812-8e259ae74d4e
Content-Type: application/json

{"accountInfo": "@anna.smith", "imageUrl": null}
```

Response: HTTP 200 → Toggle → isActive: true

Flow 3: Staff via Invite — Mike Johnson (staff2@nexora-dev.vlinkhub.com)

Step 3.1 — Get Invite Info (Anonymous)

```
GET https://nexora-dev-api.vlinkhub.com/api/v1/staff/invite/G9b-4S1zJWqvEVj145_UrCq-...
```

Response: HTTP 200

```
{"invitedName": "Mike Johnson", "invitedPosition": "Nail Artist", "businessName": "Nail Hub Dev"}
```

Step 3.2 — Signup Staff2

Response: HTTP 201 — userId: 6ebb45ae-44c5-4786-b2e1-c64f18459c1b

Step 3.3 — Sign in Staff2

Response: HTTP 200 — JWT token returned

Step 3.4 — Accept Invite (Anonymous)

Response: HTTP 204

DB verification:

- `StaffProfile.Id = b48d5643-4fde-4924-a17a-c2fa610b7c2e` , `StaffCode = NYINFFK30`

Step 3.5 — Get Payment Methods

Response: HTTP 200 — 5 methods pre-created. Zelle ID: d7e12e54-46e4-4d40-b4c5-4cbe0ee20255

Step 3.6 — Configure Zelle (mike@nexora-dev.vlinkhub.com)

```
PUT .../payment-methods/d7e12e54-46e4-4d40-b4c5-4cbe0ee20255

{"accountInfo": "mike@nexora-dev.vlinkhub.com", "imageUrl": null}
```

Response: HTTP 200 → Toggle → isActive: true

Flow 4: Customer Tip Flow (Anonymous — No Auth Required)

Step 4.1 — Get Touch Page (Customer scans QR at Table #1)

```
GET https://nexora-dev-api.vlinkhub.com/api/v1/touch/nail-hub-dev/table-1
```

```
{
  "business": { "name": "Nail Hub Dev", "logoUrl": null, "description": "Dev environment nail salon for API testing"},
  "touchPoint": { "id": "0a7c35bd-5424-4eac-8e66-05c7c97e7982", "name": "Table #1", "type": "Table", "assignedStaffProfileId": null},
  "staff": [
    { "id": "bebbdfb9-43af-4c26-a056-70b0d35aeb91", "displayName": "Anna Smith", "position": "Nail Tech", "availablePaymentMethods": [] },
    { "id": "b48d5643-4fde-4924-a17a-c2fa610b7c2e", "displayName": "Mike Johnson", "position": "Nail Artist", "availablePaymentMethods": [] }
  ]
}
```

GET https://nexora-dev-api.vlinkhub.com/api/v1/touch/payment-link?staffId=bebbdfb9-43af-4c26-a056-70b0d35aeb91&method=0&amount=5.00

```
{"redirectUrl": "https://cash.app/$anna.smith?amount=5.00", "zellePhone": null, "zelleEmail": null, "appleCashPhone": null}
```

```
POST https://nexora-dev-api.vlinkhub.com/api/v1/touch/tip
Content-Type: application/json
```

```
{
  "touchPointId": "0a7c35bd-5424-4eac-8e66-05c7c97e7982",
  "staffProfileId": "bebbdbf9-43af-4c26-a056-70b0d35aeb91",
  "amount": 5.00,
  "paymentMethod": 0
}
```

```
{"tipId": "5ae07ccc-4e79-4e1b-890b-844856cbd597"}
```

POST <https://nexora-dev-api.vlinkhub.com/api/v1/touch/tip/5ae07ccc-4e79-4e1b-890b-844856cbd597/confirm>

Step 4.5 – Create Review (5★ → Public → Google)

```
POST https://nexora-dev-api.vlinkhub.com/api/v1/touch/review
Content-Type: application/json
```

```
{
  "touchPointId": "0a7c35bd-5424-4eac-8e66-05c7c97e7982",
  "tipId": "5ae07ccc-4e79-4e1b-890b-844856cbd597",
  "staffProfileId": "bebbdfb9-43af-4c26-a056-70b0d35aeb91",
  "rating": 5,
  "comment": "Amazing service! Anna did a fantastic job!",
  "customerName": "John Doe",
  "customerEmail": "john@test.com"
}
```

Response: HTTP 201

```
{"reviewId": "cdb6f10a-5cc3-422b-892e-2b034072dc01"}
```

DB: RoutingType=Public , RoutingDestination=NULL ⚠️ (see Findings)

Step 4.6 — Track Google Click

POST https://nexora-dev-api.vlinkhub.com/api/v1/touch/review/cdb6f10a-5cc3-422b-892e-2b034072dc01/track-google

Response: HTTP 200

Step 4.7 — Create Tip (\$10 Venmo → Mike)

POST https://nexora-dev-api.vlinkhub.com/api/v1/touch/tip
Content-Type: application/json

```
{
  "touchPointId": "0a7c35bd-5424-4eac-8e66-05c7c97e7982",
  "staffProfileId": "b48d5643-4fde-4924-a17a-c2fa610b7c2e",
  "amount": 10.00,
  "paymentMethod": 1
}
```

Response: HTTP 201 — tipId: b5bd8888-0430-4706-8ddd-4af5ec5b2e84

Step 4.8 — Confirm Tip #2

Response: HTTP 200

Step 4.9 — Create Review (2★ → Private → Merchant Email)

POST https://nexora-dev-api.vlinkhub.com/api/v1/touch/review
Content-Type: application/json

```
{
  "touchPointId": "0a7c35bd-5424-4eac-8e66-05c7c97e7982",
  "tipId": "b5bd8888-0430-4706-8ddd-4af5ec5b2e84",
  "staffProfileId": "b48d5643-4fde-4924-a17a-c2fa610b7c2e",
  "rating": 2,
  "comment": "Service was okay but needs improvement in communication",
  "customerName": "Jane Doe",
  "customerEmail": "jane@test.com"
}
```

Response: HTTP 201 — reviewId: 178db669-4d09-4591-a6fb-eb91c30a2786

DB: RoutingType=Private , RoutingDestination=NULL ⚠️ (see Findings)

Step 4.10 — Skip Tip at Front Desk

POST https://nexora-dev-api.vlinkhub.com/api/v1/touch/tip/skip
Content-Type: application/json

```
{"touchPointId": "99b1295b-84d1-400e-9012-8cbdd4d35807"}
```

Response: HTTP 201

Flow 5: Merchant Dashboard Verification

Step 5.1 — Dashboard Overview

GET https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/overview
Authorization: Bearer <MERCHANT_TOKEN>

Response: HTTP 200

```
{
  "tipsSummary": {"totalAmount": 15.00, "totalCount": 2, "avgAmount": 7.50, "previousPeriodComparison": 100},
  "scansSummary": {"totalPageViews": 0, "conversionRate": 0},
  "reviewsSummary": {"totalCount": 2, "avgRating": 3.5, "count4To5Stars": 1, "count1To3Stars": 1, "googleClickCount": 1, "yelpClickCount": 1}
}
```

- ✅ totalAmount=15.00 (confirmed tips only: 5+10, skipped excluded)
- ✅ googleClickCount=1 (track-google called in Step 4.6)
- ⚠️ totalPageViews=0 — QR scans via API are not counted (frontend-only tracking)

Step 5.2 — Dashboard Overview with Date Filter

GET .../dashboard/overview?dateFrom=2026-06-07&dateTo=2026-06-07

Response: HTTP 200 — Same data (date filter works, all data is from today)

Step 5.3 — Dashboard Reviews

GET https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/reviews

Response: HTTP 200

```
{
  "items": [
    {"id": "178db669-...", "rating": 2, "staffName": "Mike Johnson", "routingType": "Private", "isResolved": false, "customerEmail": "..."},
    {"id": "cdb6f10a-...", "rating": 5, "staffName": "Anna Smith", "routingType": "Public", "googleClickedAt": "2026-06-07T07:32:51.27"}
  ],
  "totalCount": 2
}
```

Step 5.4 — Reviews Filter: Public

GET .../dashboard/reviews?routingType=Public

Response: HTTP 200 — 1 item (5★ Anna)

Step 5.5 — Reviews Filter: Private

GET .../dashboard/reviews?routingType=Private

Response: HTTP 200 — 1 item (2★ Mike)

Step 5.6 — Resolve Private Review #2

PUT https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/reviews/178db669-4d09-4591-a6fb-eb91c30a2786/resolve
Authorization: Bearer <MERCHANT_TOKEN>

Response: HTTP 204

Step 5.7 — Dashboard Tips

GET https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/tips

Response: HTTP 200

```
{
  "items": [
    {"amount": 0.00, "status": "Skipped", "touchPointName": "Front Desk", "staffName": null},
    {"amount": 10.00, "status": "Confirmed", "paymentMethod": "Venmo", "staffName": "Mike Johnson", "touchPointName": "Table #1"},
    {"amount": 5.00, "status": "Confirmed", "paymentMethod": "CashApp", "staffName": "Anna Smith", "touchPointName": "Table #1"}
  ],
  "totalCount": 3
}
```

Step 5.8 — Staff Metrics

GET https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/staff

Response: HTTP 200

```
[
  {"displayName": "Mike Johnson", "tipTotal": 10.00, "tipCount": 1, "avgRating": 2, "reviewCount": 1},
  {"displayName": "Anna Smith", "tipTotal": 5.00, "tipCount": 1, "avgRating": 5, "reviewCount": 1}
]
```

Step 5.9 — Touchpoints Metrics

GET https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/touchpoints

Response: HTTP 200

```
[
  {"name": "Anna's Card", "type": "StaffCard", "scanCount": 0, "tipCount": 0, "tipTotal": 0, "ctr": 0},
  {"name": "Front Desk", "type": "FrontDesk", "scanCount": 0, "tipCount": 0, "tipTotal": 0, "ctr": 0},
  {"name": "Table #1", "type": "Table", "scanCount": 0, "tipCount": 2, "tipTotal": 15.00, "ctr": 0, "avgRating": 3.5}
]
```

Step 5.10 — Staff List

GET https://nexora-dev-api.vlinkhub.com/api/v1/merchant/staff

Response: HTTP 200

```
[
  {"displayName": "Anna Smith", "staffCode": "N99MMUIQK", "position": "Nail Tech", "status": "Active", "tipCount": 1, "averageRating": 5},
  {"displayName": "Mike Johnson", "staffCode": "NYINFFK30", "position": "Nail Artist", "status": "Active", "tipCount": 1, "averageRating": 2}
]
```

Step 5.11 — Touchpoints List

GET <https://nexora-dev-api.vlinkhub.com/api/v1/merchant/touchpoints>

Response: HTTP 200 — 3 active touch points:

- annas-card (StaffCard → Anna) — URL: <https://nexora.vlinkpay.com/touch/nail-hub-dev/annas-card>
- front-desk (FrontDesk)
- table-1 (Table)

Step 5.12 — Business Profile

GET <https://nexora-dev-api.vlinkhub.com/api/v1/merchant/business>

Response: HTTP 200

```
{
  "id": "e734f8db-5dad-4233-a7a6-039daccf92dd",
  "name": "Nail Hub Dev",
  "slug": "nail-hub-dev",
  "isPublic": true,
  "onboardingStep": 5,
  "onboardingCompletedAt": "2026-06-07T07:30:34.729741",
  "subscription": {"plan": "Starter", "status": "Trialing", "trialEndsAt": "2026-07-07T07:27:12.735622"}
}
```

Step 5.13 — Tips Chart (US-11)

GET <https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/tips-chart?groupBy=day>

Response: HTTP 404 — Endpoint not yet implemented.

Step 5.14 — Top Staff (US-11)

GET <https://nexora-dev-api.vlinkhub.com/api/v1/merchant/dashboard/top-staff?limit=3>

Response: HTTP 404 — Endpoint not yet implemented.

Flow 6: Public Endpoints

Step 6.1 — Touch Page (Anna's Card — StaffCard)

GET <https://nexora-dev-api.vlinkhub.com/api/v1/touch/nail-hub-dev/annas-card>

Response: HTTP 200 — Shows only Anna with CashApp + Venmo options (Mike not in list — StaffCard filter works )

Step 6.2 — Business Payment Methods (Public)

GET <https://nexora-dev-api.vlinkhub.com/api/v1/public/businesses/e734f8db-5dad-4233-a7a6-039daccf92dd/payment-methods>

Response: HTTP 200 — [] (no business-level payment methods configured)

Step 6.3 — Active Banners

GET https://nexora-dev-api.vlinkhub.com/api/v1/banners/active

Response: HTTP 200 — [] (no banners seeded in test DB)

Step 6.4 — Blog Categories

GET https://nexora-dev-api.vlinkhub.com/api/v1/blog-categories

Response: HTTP 200 — [] (test DB has no seeded blog categories — differs from local dev which had did-you-know)

Step 6.5 — Blogs (Known Bug)

GET https://nexora-dev-api.vlinkhub.com/api/v1/blogs

Response: HTTP 404 — Confirmed [NonController] attribute on PublicBlogsController disables this endpoint.

Step 6.6 — Staff Tip History

GET https://nexora-dev-api.vlinkhub.com/api/v1/staff/tips

Authorization: Bearer <STAFF1_TOKEN>

Response: HTTP 200

```
{
  "items": [
    {
      "id": "5ae07ccc-4e79-4e1b-890b-844856cbd597",
      "amount": 5.00,
      "status": "Confirmed",
      "paymentMethod": "CashApp",
      "touchPointName": "Table #1",
      "businessName": "Nail Hub Dev",
      "confirmedAt": "2026-06-07T07:32:50.590175"
    }
  ],
  "totalCount": 1
}
```

Flow 7: Staff Management API

Step 7.1 — Toggle Mike → Inactive

PUT https://nexora-dev-api.vlinkhub.com/api/v1/merchant/staff/ed492187-de43-47c4-910a-a2a4e8d6e80d/status

Authorization: Bearer <MERCHANT_TOKEN>

Content-Type: application/json

```
{"status": "Inactive"}
```

Response: HTTP 204

Step 7.1b — Verify Mike not on Touch Page

GET https://nexora-dev-api.vlinkhub.com/api/v1/touch/nail-hub-dev/table-1

Response: HTTP 200 — Only Anna in staff list (count: 1) 

Step 7.2 — Toggle Mike back → Active

```
PUT .../staff/ed492187-de43-47c4-910a-a2a4e8d6e80d/status

{"status": "Active"}
```

Response: HTTP 204

Step 7.3 — Reorder Staff

```
PUT https://nexora-dev-api.vlinkhub.com/api/v1/merchant/staff/reorder
Authorization: Bearer <MERCHANT_TOKEN>
Content-Type: application/json

[{"id": "ed492187-...", "sortOrder": 0}, {"id": "56f9cefe-...", "sortOrder": 1}]
```

Response: HTTP 500  — COMMON_INTERNAL_SERVER_ERROR (also failed with staffProfileId key format)

Step 7.4 — Resend Invite for Active Staff (should fail)

```
POST https://nexora-dev-api.vlinkhub.com/api/v1/merchant/staff/56f9cefe-5200-473e-9456-93ed293f6cfa/resend-invite
Authorization: Bearer <MERCHANT_TOKEN>
```

Response: HTTP 405 — Method Not Allowed. Endpoint may not be implemented or route mismatch.

Flow 8: Security Tests

Test	Endpoint	Token	HTTP Status
SEC-01a: IDOR dashboard	GET /api/v1/merchant/dashboard/overview	STAFF1_TOKEN	403 — USER_NOT_MERCHANT
SEC-01b: IDOR tips	GET /api/v1/merchant/dashboard/tips	STAFF1_TOKEN	403 — USER_NOT_MERCHANT
SEC-02: Staff detail via merchant	GET /api/v1/merchant/staff/<STAFF1_PROFILE_ID>	MERCHANT_TOKEN	404 — endpoint doesn't exist
SEC-02b: Staff list payment handle exposure	GET /api/v1/merchant/staff	MERCHANT_TOKEN	Response keys: id, displayName, position, status, tipCount, average — no payment fields
SEC-03a: Touch page no auth	GET /api/v1/touch/nail-hub-dev/table-1	None	200
SEC-04: Token reuse	POST /staff/invite/<ACCEPTED_TOKEN>/accept	None	400 — STAFF_INVITE_ALREADY_ACCEPTED

Test	Endpoint	Token	HTTP Status
SEC-05: Starter plan gate	POST /api/v1/merchant/touchpoints (4th)	MERCHANT_TOKEN	400 — TOUCHPOINT_STARTER_LIMIT_REACHED — message đạt giới hạn 3 touch points của gói Starter. Nâng cấp lên tiếp tục."
SEC-06: Staff tries merchant endpoints	PUT /merchant/business/review-links	STAFF1_TOKEN	403 — USER_NOT_MERCHANT

Acceptance Criteria Results

AC	Description	Status	Notes
AC-01	Merchant onboard 5 steps	✔ Pass	isPublic=true , onboardingStep=5
AC-01	Slug unique check	✔ Pass	Slug nail-hub-dev created successfully
AC-02	Staff invite flow	✔ Pass	StaffInvite.Status=Accepted , BusinessStaffLink.Status=Active
AC-02	Staff multi-business	☐ Not tested	Requires second merchant account
AC-02	Payment handles hidden from merchant	✔ Pass	Staff list: no cashApp/venmo/zelle fields
AC-03	Touch page accessible after onboarding	✔ Pass	GET /touch/nail-hub-dev/table-1 → 200
AC-03	QR URL format	✔ Pass	URL: https://nexora.vlinkpay.com/touch/nail-hub-dev/table-1
AC-04	Tip flow no login required	✔ Pass	All touch/tip endpoints anonymous
AC-04	Bỏ qua tip (skip)	✔ Pass	Tip.Status=Skipped
AC-05	4-5★ → Public routing	✔ Pass	RoutingType=Public
AC-05	1-3★ → Private routing	✔ Pass	RoutingType=Private
AC-05	Google click tracking	✔ Pass	googleClickedAt populated after track-google
AC-06	Dashboard metrics correct	✔ Pass	totalAmount=15.00 (confirmed only), avgRating=3.5
AC-06	Date filter works	✔ Pass	?dateFrom=&dateTo= filters correctly
AC-06	Resolve private feedback	✔ Pass	PUT .../resolve → 204
AC-06	IDOR — merchant sees only own data	✔ Pass	Staff token gets 403
AC-07	Starter limit 3 touchpoints	✔ Pass	4th → 400 TOUCHPOINT_STARTER_LIMIT_REACHED
SEC-04	Token single use	✔ Pass	Reuse → 400 STAFF_INVITE_ALREADY_ACCEPTED

Observations & Findings

Issues Found

#	Endpoint	Issue	Severity
1	GET /api/v1/blogs	[NonController] attribute on PublicBlogsController — endpoint không đăng ký với router → 404	🔴 Bug (confirmed, same as local)

#	Endpoint	Issue	Severity
2	PUT /api/v1/merchant/staff/reorder	Trả về 500 COMMON_INTERNAL_SERVER_ERROR với cả hai format body (id và staffProfileId)	🔴 Bug (mới phát hiện)
3	POST /api/v1/merchant/staff/{id}/resend-invite	Trả về 405 Method Not Allowed — endpoint chưa implement hoặc route không khớp	🔴 Bug (mới phát hiện)
4	GET /swagger và GET /swagger/index.html	Swagger không accessible dù EnableSwagger: true trong appsettings.Test.json — bị block ở nginx/reverse proxy	⚠️ Cấu hình
5	Review.RoutingDestination	Cột RoutingDestination là NULL trong DB cho cả Public và Private reviews — URL Google/Yelp và email không được lưu	⚠️ Cần điều tra — routing destination có thể được resolve dynamically từ Business record thay vì lưu trực tiếp
6	GET /api/v1/merchant/dashboard/tips-chart	404 — chưa implement (US-11 spec)	⚠️ Missing endpoint
7	GET /api/v1/merchant/dashboard/top-staff	404 — chưa implement (US-11 spec)	⚠️ Missing endpoint
8	GET /api/v1/blog-categories	Trả về [] — test DB chưa được seed với blog categories (local dev có did-you-know)	⚠️ DB seeding
9	Invalid invite token	Trả về 404 thay vì 410 Gone như spec US-13 yêu cầu	⚠️ HTTP code không đúng spec
10	scansSummary.totalPageViews	Luôn = 0 — scan count chỉ được track qua frontend, không qua API	⚠️ Expected behavior

API Route Corrections (vs. Plan)

Wrong Route	Correct Route
Merchant staff detail	GET /api/v1/merchant/staff/{staffProfileId} → 404 (không có endpoint này)
PUT /merchant/staff/{id}/status	Dùng BusinessStaffLink ID (không phải StaffProfile ID)

Key Behavior Notes

- **Signup auto-activates:** IsEmailVerified=true , Status=Active ngay sau signup — không cần xác thực email.
- **Payment methods pre-created:** AcceptInviteCommand tự động tạo 5 StaffPaymentMethod records với isActive=false . Dùng PUT để update + PATCH /toggle để kích hoạt.
- **Invite token not returned by API:** InviteStaffCommand chỉ trả về inviteId . Token phải lấy từ DB (StaffInvites."Token").
- **Touch point slug auto-generated:** Hệ thống tự tạo từ name (e.g., "Table #1" → "table-1").
- **Review routing:** 4-5★ → RoutingType=Public , 1-3★ → RoutingType=Private . Logic trong CreateReviewCommandHandler .
- **Touch page URL:** https://nexora.vlinkpay.com/touch/{slug}/{touchpoint-slug} (base URL từ appsettings.json , không phải dev URL).
- **Staff toggle status:** Dùng BusinessStaffLink ID (trường id trong staff list, không phải staffProfileId).
- **Onboarding required:** Business phải gọi POST /merchant/business/complete-onboarding trước khi touch page accessible công khai.
- **Dashboard fixes:** So với local testing (trả về 0 cho mọi metric), dev environment trả về metrics chính xác — bug local đã được resolve.